

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 2.3: Excretion in Plants — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 2.3: Excretion in Plants
Driving Question	DRIVING QUESTION: How do trees and crops clear out their metabolic wastes without any organs or excretory systems, and why are these waste products worth so much money in our markets?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Where Do Plant Wastes Go? Gum, Droplets and Falling Leaves	Teacher note: Learners observe the anchoring phenomena, namely gum oozing from an acacia or gum arabic tree, latex from a cut euphorbia or pawpaw, guttation droplets on maize or sukuma wiki leaf edges at dawn, and fallen leaves under a tree. Use real specimens or photographs. Before showing them, give 2 minutes of WAIT TIME for learners to predict: 'Where do plants put their wastes if they have no kidneys?' Cold-call 4 learners and record the predictions on the board. Look for the misconception that plants do not produce wastes at all.	Excretion is the removal of metabolic wastes from the body. Plants lack complex excretory organs such as kidneys because (1) their metabolic rate is low, so they produce small amounts of waste, (2) the gaseous wastes (oxygen from photosynthesis, carbon dioxide from respiration) are recycled or easily diffused out, and (3) they can store many wastes in dead or non-vital tissues. Teacher note: stress that excretion differs from egestion and secretion. Ask learners to link this to ugali-grade maize growing without any 'excretory system'.	Teacher answer key: Plants do not need kidneys because they produce fewer toxic wastes, recycle gases between photosynthesis and respiration, and store or shed wastes safely. Learners should be able to explain that the gum, latex, droplets and fallen leaves are all evidence of waste removal or waste storage. Add the first questions to the Driving Question Board, for example 'Why is gum arabic valuable?' and start the initial model with a simple sketch of a plant showing 'wastes leave' and 'wastes stored'. Accept incomplete explanations at this stage and flag them for revision.
Lesson 2 How Do Plants Get Rid of Wastes? Nature Walk, Exudation and Guttation Evidence	Teacher note: Learners go on a nature walk around the school compound, or use photographs, to find evidence of waste removal. They look for stomata and lenticels (use a hand lens on leaf peels and on mature stems such as those of a guava or Grevillea tree), water droplets, gum on acacia, latex from a cut pawpaw or euphorbia, and resin on a cypress or pine tree. Give each group a recording table with columns for the observation, the waste, and	Plants eliminate gaseous and liquid wastes by diffusion through stomata and lenticels, transpiration (water vapour), exudation (gums, resins, latex) and guttation (water and dissolved wastes from hydathodes). Teacher note: distinguish guttation (liquid, at night or early morning, high humidity, through hydathodes) from transpiration (water vapour, mostly in the day, through stomata). Oxygen leaves through stomata during the day, and carbon dioxide leaves	Teacher answer key: Plants use diffusion for gases, transpiration for water vapour, exudation for gums, resins and latex, and guttation for liquid wastes. Learners link each observation from the walk to its route, for example the latex from a cut pawpaw to exudation and the droplets on maize leaf margins to guttation. Prompt: 'Why can plants rely on diffusion when animals need blood transport?' Because plants have low metabolic rates and large surface areas.

	the route of removal. Cold-call 3 groups for evidence and use WAIT TIME of 5 seconds before they answer.	during respiration at night.	Add new questions to the DQB such as 'What happens to wastes that cannot leave?' and revise the model by adding arrows for each route.
Lesson 3 Where Do the Wastes Go? Shedding Old Leaves and Locking Away Crystals	Teacher note: Learners examine old and young leaves, bark, fruit peels and flowers from local plants such as tea in Kericho, mango, avocado and sisal. Give a practical or a photographic demonstration. Learners may view calcium oxalate crystals (raphides) in a sliced Dieffenbachia, sisal or spinach-type leaf under a microscope, if available. They compare the colour and texture of old leaves with those of young leaves and notice the tannin-stained bark. Allow 3 minutes of silent observation, then ask: 'Why do old leaves fall off?'	Plants deposit insoluble metabolic wastes such as calcium oxalate crystals, tannins and resins in non-vital organs (bark, old leaves, fruits and flowers) or in vacuoles. These wastes are removed when the organs are shed, for example in leaf fall, bark peeling and the dropping of fruits and petals. Teacher note: stress that this storage makes the wastes harmless and that deciduous trees in the Rift Valley shed leaves in the dry season, which also reduces water loss.	Teacher answer key: Plants lock harmful wastes in insoluble forms (crystals and tannins) in vacuoles and in dead or non-vital tissues. When the leaf, bark or fruit is shed, the wastes leave the plant, and so plants do not need excretory organs. Learners should explain why storage in dead tissues is safe: the wastes are insoluble and therefore cannot disrupt metabolism. Update the DQB with questions such as 'Which stored wastes do people use?' and add shedding and crystal storage to the cumulative model.
Lesson 4 Cash from Waste: Categorising Plant Excretory Products and Their Value in Kenya	Teacher note: Learners handle or view samples or pictures of products from plant wastes: latex (rubber, papain from pawpaw), tannins (tea from Kericho, wattle bark for leather tanning), alkaloids (caffeine in coffee from Nyeri, quinine, nicotine from tobacco in Migori, and the pyrethrum used in insecticides), gums and resins (gum arabic, myrrh and frankincense from northern Kenya), and essential oils (eucalyptus, lemon grass). In groups, learners sort the samples onto a chart. Use a jigsaw activity in which each group becomes expert on one product and teaches the others. Use cold-calls, with 3 learners per group.	Plant excretory products fall into types: latex, tannins, alkaloids, gums and resins, and essential oils. Each comes from particular plant parts and has particular uses. Latex is used to make rubber and papain, tannins in tanning leather, tea and dyes, alkaloids in medicines, stimulants and pesticides, gums and resins in adhesives, food and incense, and essential oils in perfumes and medicines. Teacher note: ask learners to link each product to the region where it is grown in Kenya and to its value as an income source.	Teacher answer key: These wastes are valuable because they have useful chemical properties that people can use in industry, medicine and food, and so farmers earn income from products that the plant itself treats as wastes. Learners should name at least one product per category, its source plant and its use or market value, such as coffee and tea as export earners. This answers the second part of the driving question. Add new questions to the DQB, for example 'Are all plant wastes safe to use?' and update the model with a labelled 'products and uses' section.
Lesson 5 Who Profits from Plant Wastes? Community Uses, Alkaloid Risks and the Final Explanation	Teacher note: Learners review a case study or a local interview about a community that earns income from plant wastes, such as gum and resin collectors in Turkana or Marsabit, tea pickers in Kericho, pyrethrum farmers in Nakuru or sisal growers. They also look at evidence about the risks of alkaloids, such as nicotine in tobacco, caffeine misuse, miraa (khat) and poisonous plants for livestock. In groups, learners discuss the benefits and risks and use WAIT TIME of 10 seconds before they	Plants lack excretory organs because they produce few toxic wastes, recycle gases, and store wastes in dead tissues, and these wastes lead to valuable products that support livelihoods. Alkaloids can be harmful, as nicotine and caffeine can be addictive and some plant wastes are poisonous, so they must be used responsibly. Teacher note: connect this to values and PCIs such as health education, responsibility and economic opportunity, and encourage learners to suggest safe	Teacher answer key: Learners write the final explanation for the driving question. Trees and crops clear their wastes without organs by diffusion of gases through stomata and lenticels, by transpiration, exudation and guttation, and by storing insoluble wastes in non-vital tissues that are later shed. These wastes are worth money because they provide rubber, tea, coffee, medicines, gums, resins and oils, though harmful alkaloids need careful use. Check that the final model includes all the routes,

	report. Ask: 'Who benefits, and who can be harmed?'	and sustainable harvesting such as tapping gum without killing the tree.	the storage sites and the product uses, and that learners revise their earlier predictions from Lesson 1 and answer the questions on the DQB.
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